

Hello everyone,

My name is Zac Thompson. I am currently a music teacher, as well as a husband and a dad to a 3-month-old daughter. When I have free time outside of teaching, studying, grading, and family life, I enjoy woodworking, playing tabletop RPGs like Dungeons & Dragons and Daggerheart, video games, and working on coding projects that support both my classroom and personal interests.

Number Theory in Real-World Applications

Number theory shows up in more places than I expected, especially in computer science. One of the most prominent examples is encryption, where prime numbers and modular arithmetic are used to securely transmit information such as passwords and credit card data over the internet. It also plays a role in hashing and error detection, which help ensure data stays accurate and secure when systems communicate (Patel, 2025).

Differentiation Between Discrete and Continuous Numbers

Number theory focuses on discrete numbers, which are values that can be counted and distinguished, such as integers. A simple example would be counting the number of people in a stadium. Continuous values, on the other hand, can take on infinitely precise values, such as temperature or distance. So, while continuous data exists over a smooth range, discrete values consist of distinct points, which makes them easier to work with in computational systems (Chapple, 2021).

Role of Discrete Numbers

Discrete numbers form the foundation of number theory because they allow us to study concepts such as divisibility, prime numbers, and relationships among values in a precise and

structured way. While continuous numbers are important in other areas of mathematics, in practice, they are often converted to discrete values through rounding or grouping to enable more effective analysis. This stood out to me because it shows that even when data is theoretically continuous, it is frequently represented in a discrete form due to measurement limitations and the need for simpler computation.

Discrete vs. Continuous in Number Theory

Number theory helps make the distinction clearer because algorithms rely on discrete inputs to function. Even when working with continuous data, such as time or measurements, it is often rounded, grouped, or sampled into discrete values before analysis. This reinforces the idea that computers fundamentally operate on discrete structures, even when modeling more complex, continuous systems (Kazlauskas, 2021).

Best Regards,

Zac Thompson

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